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Photobiomodulation in Treatment of Head and Neck Cancer-Related Chronic Lymphedema: A Pilot Randomized Controlled Trial

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Abstract

Background: Chronic lymphedema is a common late effect after completion of head and neck cancer (HNC) treatment, contributing to substantial symptom burden and negatively impacting quality of life. No effective approaches are available to treat this progressive condition. This study aimed to evaluate the preliminary efficacy of photobiomodulation (PBM) therapy for chronic lymphedema in HNC survivors.

Methods: This was a pilot, randomized, wait-list controlled trial. Eligible HNC survivors included those with chronic lymphedema after completion of complete decongestive therapy. Participants were randomized (1:1) to 1 of 2 arms: intervention group (active) or wait-list control group (control). The active group received 12 PBM therapy sessions (twice a week for 6 weeks). The control group completed the study assessments and was then offered the same dose of PBM therapy as the active group. Lymphedema and fibrosis (LEF), symptom burden, jaw range of motion, and neck range of motion were measured at baseline, end-of-intervention, 4-week, and 8-week post-intervention.

Results: Twenty five HNC survivors were randomized to the active group (n = 12) and the control group (n = 13). About 91.7% planned PBM treatment sessions were completed. No adverse events were reported. Compared to the control group, the active group demonstrated improvements at 8-week post-intervention in the severity of external LEF ($P < .001$), symptom burden (eg, Soft Tissue and Neurologic Toxicity subscale, Cohen's $d = -0.47$), and neck range of motion (eg, extension, Cohen's $d = 0.65$).

Conclusion: PBM therapy may improve chronic lymphedema-associated outcomes. Future large randomized controlled trials are warranted to examine the efficacy of PBM therapy for HNC-related chronic lymphedema.

Keywords

head and neck cancer, chronic lymphedema, photobiomodulation (PBM), survivorship

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Introduction

In 2026, an estimated 72 770 Americans will develop head and neck cancer (HNC).[1] The head and neck contains an extensive network of lymphatic channels and >300 lymph nodes (1/3 of the lymph nodes in the body).[2] HNC and its treatment often disrupt lymphatic structures and surrounding soft tissue, blocking lymph flow and causing fluid build-up in the soft tissue.[3,4] Thus, HNC survivors are at high risk for developing lymphedema, with greater than three-quarters of HNC survivors developing lymphedema by 3 months

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post-cancer treatment.[5] Head and neck lymphedema can occur both externally and internally.[3-7] Involvement of external sites may lead to skin tightness and pain,[5-8] while involvement of internal structures results in functional deficits such as difficulty chewing and speaking.[9,10] Intra-oral and pharyngeal edema can cause decreased swallowing safety and efficiency, and may necessitate a gastrostomy tube for feeding.[10]

The cornerstone of standard therapy for head and neck lymphedema is Complete Decongestive Therapy (CDT) provided by a certified lymphedema therapist.[11] Therapists use manual lymphatic drainage, compression garments, exercise, and skin care to reduce visible or palpable swelling and reduce fibrosis.[2,11,12] After completion of lymphedema therapy, some patients still have clinically evident residual lymphedema that can progress to chronic phase lymphedema.[8,13] Currently, there are no proven effective therapies for treatment of refractory or chronic lymphedema in HNC survivors.[14] While several pharmacological interventions are under investigation for fibrotic processes such as chronic lymphedema and fibrosis (eg, vitamin E, trental, and statins), level 1 evidence demonstrating efficacy is lacking.[14] Innovative solutions are therefore needed to manage chronic lymphedema in HNC survivors. One potential solution is photobiomodulation (PBM) therapy, previously known as low-level laser therapy.[15]

PBM is defined as the use of "non-ionizing optical radiation in the visible and near-infrared (NIR) spectral range, which is absorbed by endogenous chromophores to elicit photophysical and photochemical events at various biological scales without eliciting thermal damage and leading to physiological changes and therapeutic benefits."[15] PBM has been investigated for therapeutic purposes for more than 40 years.[16] Animal studies have shown that PBM reduces lymphostatic fibrosis, stimulates lymphangiogenesis, and enhances lymphatic motility,[17-21] supporting the potential for PBM in treating chronic lymphedema.

Three systematic reviews[22-24] and one meta-analysis[25] of 16 clinical trials (12 randomized) found that: (1) PBM reduced arm swelling and pain, and improved quality of life (QOL) among breast cancer survivors; (2) extended PBM therapy course (>15 treatment sessions) yielded better outcomes;[25] (3) PBM generated a long-term effect on arm lymphedema showing benefit at 12-month post PBM therapy;[26,27] and (4) no adverse events were identified from PBM therapy. The U.S. Food and Drug Administration (FDA) approved PBM as a treatment approach for breast cancer-related arm lymphedema in 2006.[28] Given the similarities of underlying pathobiological changes between breast cancer-related lymphedema and HNC-related lymphedema (eg, lymphatic system structural damage,[2,14] and accumulation of lymphatic fluid in interstitial tissues[2,14]), it would be beneficial to determine the effectiveness of PBM therapy for head and neck chronic lymphedema.

Currently, only case studies indicate the potential value of using PBM therapy for treating head and neck lymphedema.[29,30] No randomized clinical trials have examined the effect of PBM therapy on head and neck cancer-related chronic lymphedema. We completed a pilot study using a pre-post design that demonstrated the feasibility, acceptability, and potential efficacy of PBM for this indication.[31,32] Based on those results, we conducted a pilot randomized wait-list controlled trial to provide additional preliminary efficacy data of PBM for treating chronic lymphedema in HNC survivors, the results of which could inform future study design. The primary aim of our study was to determine the impact of PBM therapy, as compared to a wait-list control, on changes in the severity of lymphedema (primary outcome measure). The secondary aim was to determine the impact of PBM therapy, as compared to a wait-list control, on lymphedema-related symptom burden and functional impairments (secondary outcome measures). Here, we report the results.

Methods

Design

This pilot randomized, wait-list controlled trial (ClinicalTrials.gov identifier: NCT04482855) was conducted at the Abramson Cancer Center of the University of Pennsylvania. The study (UPCC 01320) was approved by the Institutional Review Board and Clinical Trial Scientific Research Monitoring Committee. The study was conducted per the Declaration of Helsinki and the International Conference on Harmonization guidelines for Good Clinical Practice. All participants signed informed consent forms before participating in any study activities.

Patient Selection

Participants were recruited and enrolled from February 2020 to July 2022. Inclusion criteria were >18 years of age; >6 months post HNC treatment with no documented radiographic or clinical evidence of recurrent or progressive cancer; having head and neck lymphedema with or without fibrosis; completion of initial lymphedema therapy; lymphedema duration between 3 and 36 months; ability to speak and read English; ability to provide informed consent; fully vaccinated against COVID-19 (or for unvaccinated patients, completing a weekly screening COVID-19 test during the 6-week intervention). Exclusion criteria included any of the following medical conditions that would prohibit the safe implementation of PBM therapy: pregnancy; acute infection; photosensitivity; chronic inflammatory diseases; venous thrombosis; carotid artery stenosis; history of severe trauma; use of medication that

affects body fluid and electrolyte balance; use of high doses of non-steroidal anti-inflammatory drugs; pre-existing skin rash, ulceration, or open wound in the treatment area; active lymphedema therapy or physical therapy; and allergic and other systemic skin diseases.

Procedures

To ensure fidelity of data collection, all data collectors were carefully trained in the study protocol.

Interested patients were scheduled for a screening visit at the study site. An electronic consent form was obtained from all study participants before initiating any study-related research activities. Once potential participants passed screening and were enrolled in the trial, they were scheduled for the baseline study visit. After completion of the baseline measures, participants were randomized and assigned (1:1) to 1 of 2 arms: intervention group (active) and wait-list control group (control) via a concealed allocation process. The study statistician conducted randomization via a computer-generated, permuted block program.

Participants in the active group were scheduled for the PBM therapy. After completing the 6-week PBM intervention, participants underwent another study assessment and were scheduled for the 4-week and 8-week post-intervention visits. Participants in the control group completed the same study assessments and were offered the same 6-week PBM as the active group. Participants received compensation of $135 in total for completing all 4 study assessment visits (baseline, end of intervention, 4-week, and 8-week post-intervention) during the study.

Study Intervention: Photobiomodulation (PBM) Therapy

Study Device. The RianCorp LTU-904 laser therapy unit was used as a PBM therapy device in the trial. The LTU-904 is an FDA-approved, Class I, portable laser therapy unit for treating post-mastectomy lymphedema.[28] It is a low-output laser in the infrared wavelength (904 nm), requiring no safety glasses, given no risk of eye damage. We used the following parameters that were successfully tested and finalized through our pilot study:[31] 904 nm wavelength in superpulsed mode, 5 W peak power, 5 mW average output, spot size of 0.2 cm^2, and energy density of 1.5 J/cm^2. PBM therapy was administered by a certified lymphedema therapist trained to use the LTU-904 laser therapy unit.

Study Protocol

The PBM therapy protocol was developed, tested, and finalized based on our previous study,[31] incorporating insights from patients, therapists, and oncologic clinicians. Before receiving PBM therapy, participants received 5 minutes of simple manual lymphatic drainage (MLD) to the venous angles and lateral neck bilaterally in a supine position. The MLD was administered by a certified lymphedema therapist (CLT) and followed the international standards.[2] After completion of simple MLD, participants received the PBM therapy. All participants received PBM therapy twice a week for 6 weeks (12 total sessions, with each session lasting approximately 20-30 minutes). The anatomical locations for PBM therapy were determined based on the presentation of chronic lymphedema. Based on our previous protocol, participants received 14 to 25 PBM therapy points on the face and neck area, including the maxillary prominence (1 point), mandible (2 points), preauricular (1 point), submental (3 points), sternocleidomastoid muscle (3 points), supraclavicular area (2 points), and scalene muscle (2 points).[31] Due to COVID, all participants wore a mask during the PBM therapy, impacting PBM treatment on the maxillary prominence (1 point) and preauricular (1 point). Therefore, 12 to 21 points on the face and neck were treated in this study. Each treatment point received PBM therapy for 60 seconds.[31]

Study Measures

Baseline Characteristics

Demographic Form. Participants' demographic information (eg, race and ethnicity) was collected.

Head and Neck Cancer Clinical Form. Participants' HNC (AJCC eighth edition cancer staging) and its treatment information (eg, tumor location and type of HNC treatment) were collected.

Feasibility Measures

Recruitment Log. Trained research staff documented the recruitment effort, including the number of participants recruited, consented, and screened.

PBM Implementation Log. The certified lymphedema therapist recorded the number of PBM therapy sessions, barriers to implementing PBM therapy sessions, and reasons for missed sessions.

Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. The trained study staff used this system to assess and document adverse events of the trial.[33]

Efficacy Measures. The following tools were used to collect patients' outcomes at 4 time points (ie, baseline, end of the intervention, 4- and 8-week post-intervention).

Lymphedema

Head and Neck – External Lymphedema and Fibrosis (HN-LEF) Assessment Criteria. The trained study staff used this validated tool to assess participants' head and neck external lymphedema and fibrosis (LEF) status through a physical examination. Per this tool, the anatomical sites of LEF include the left and right peri-orbital region, left and right cheeks, left and right neck, left and right supraclavicular region, and the submental area. For each anatomical site, the severity of external LEF includes no LEF (=0), mild (=1), moderate (=2), and severe (=3). For each participant, the total severity score is the sum of the severity scores of the 9 anatomical sites, which results in a score ranging from 0 (no sites with abnormalities) to 27 (all sites with severe abnormalities). The tool has good content validity, interrater reliability (84.9% agreement for grade of LEF, kappa = 0.70, $P < .001$), and excellent intrarater reliability (91.4% agreement for grade of LEF).[34,35]

Modified Patterson Scale. Internal lymphedema was scored using the Modified Patterson Scale, which documents internal swelling in the oral cavity, pharynx, and larynx. Four grades are used to rate edema level (normal = 0, mild = 1, moderate = 2, and severe = 3) for each anatomical structure. Sites are marked "N/A" when they are unable to be evaluated. The scale has good intrarater reliability (weighted kappa, 0.84) and moderate interrater reliability (weighted kappa, 0.54).[36] Due to COVID, assessment of internal lymphedema was not feasible due to exam limitations.

Symptoms

To limit the burden, patient-reported outcome measures used in this study specifically targeted symptom burden associated with head and neck lymphedema.

Head and Neck – Lymphedema and Fibrosis (HN-LEF) Symptom Inventory – Version 1.0. This validated 33-item tool was developed to measure symptom burden related to lymphedema and fibrosis in HNC patients. Each item is answered in a "yes" or "no" format. If patients answer "yes" to any given item, they are then asked to rate the intensity of the respective symptom on a 5-point scale from 1 (slight) to 5 (severe). It includes 7 subscales with good reliability, as demonstrated by an internal consistency of close to or greater than .70 (Cronbach's alpha). The 7 subscales are soft tissue and neurologic toxicity, systemic symptoms and social functioning, jaw and oral dysfunction, swallowing and taste changes, body image and sexuality, communication, and mucosal irritation.[37,38]

Neck Disability Index (NDI). This is a validated tool (10-item) designed to measure neck-specific pain and its impact on activities of daily living, including personal care, lifting, reading, headaches, concentration, work, driving, sleeping, and recreation. Each item is scored from 0 to 5. The maximum score is 50. Higher scores represent greater disability: 0 to 4 = no disability; 5 to 14 = mild disability; 15 to 24 = moderate disability; 25 to 34 = severe disability; and above 34 = complete disability. The internal consistency of the tool is adequate (Cronbach's alpha: .89-.92).[39]

Jaw and Neck Range of Motion

Jaw Range of Motion (ROM) Scale. The Jaw ROM scale is a valid and reliable tool for measuring maximal mouth opening in millimeters.[40]

Cervical Range of Motion (CROM) Device. The CROM device (Performance Attainment Associates, Lindstrom, MN) is a valid and reliable tool for measuring the amount of neck movement for the following neck movements: forward flexion, extension, right and left lateral flexion, and right and left rotation.[41]

Sample Size and Power Determination

The primary purpose of this study was to garner preliminary efficacy data of PBM therapy in HNC survivors with chronic lymphedema. The primary outcome was effect sizes. Our sample size of 12 subjects per study group[42] provided adequate initial data to estimate central tendency and variability of primary outcome values and to generate effect size estimates for efficacy outcomes.

Statistical Analysis

Participant characteristics were summarized using means and standard deviations for continuous measures, as well as frequencies and percentages for categorical measures. Stratified summaries were also prepared by the study group. The associations between group and participants' clinical and demographic characteristics were evaluated using 2-sample *t*-tests and chi-squared tests for continuous and categorical measures, respectively. Outcome measures at specific timepoints were summarized as means and standard deviations. For each outcome, a linear mixed effects model was constructed, which included a treatment group-by-time interaction. Each model employed an unstructured covariance matrix to account for the correlation between participants' repeated measures. To assess the efficacy of the intervention, the between-group difference in 8-week change from baseline was evaluated and reported for each outcome. Results from linear mixed effects models were presented as means, standard errors, and 95% confidence intervals. Because the present study was not powered to detect the statistically significant difference for certain outcomes, Cohen's *d* was computed for each outcome to provide an estimate of the standardized effect size, which

Table 1. Participant Characteristics.

Characteristics		Total (n = 25)	Intervention (n = 12)	Control (n = 13)	*P*-value
Age (y)	Mean (SD)	61.0 (5.7)	59.5 (5.1)	62.3 (6.0)	.222
	Min	51.3	52.6	51.3	
	Max	74	69	74	
Sex	Female	3 (12.0%)	2 (16.7%)	1 (7.7%)	.593
	Male	22 (88.0%)	10 (83.3%)	12 (92.3%)	
Race	Black or African American	1 (4.0%)	0 (0%)	1 (7.7%)	1.000
	White	24 (96.0%)	12 (100.0%)	12 (92.3%)	
Ethnicity	Not Hispanic	24 (96.0%)	12 (100.0%)	12 (92.3%)	1.000
	Hispanic	1 (4.0%)	0 (0%)	1 (7.7%)	
Ever used or smoked tobacco	Never	10 (40.0%)	3 (25.0%)	7 (53.8%)	.226
	Yes	15 (60.0%)	9 (75.0%)	6 (46.2%)	
Ever drink alcohol	Yes	25 (100.0%)	12 (100.0%)	13 (100.0%)	
Annual household income	$20 001 to $30 000	2 (8.0%)	0 (0%)	2 (15.4%)	.593
	$50 001 to $60 000	1 (4.0%)	1 (8.3%)	0 (0%)	
	Over $60 000	19 (76.0%)	9 (75.0%)	10 (76.9%)	
	Do not care to respond	3 (12.0%)	2 (16.7%)	1 (7.7%)	
Primary tumor location	Oral cavity	2 (8.0%)	0 (0%)	2 (15.4%)	.344
	Oropharynx	22 (88.0%)	12 (100.0%)	10 (76.9%)	
	Hypopharynx	1 (4.0%)	0 (0%)	1 (7.7%)	
Tumor-expressed virus	No known virus	3 (12.0%)	0 (0%)	3 (23.1%)	.220
	HPV	22 (88.0%)	12 (100.0%)	10 (76.9%)	
Tumor stage (TNM) at diagnosis[a]	X/Unknown	3 (12.5%)	2 (16.7%)	1 (8.3%)	1.000
	I	9 (37.5%)	4 (33.3%)	5 (41.7%)	
	II	9 (37.5%)	5 (41.7%)	4 (33.3%)	
	IVa	2 (8.3%)	1 (8.3%)	1 (8.3%)	
	IVb	1 (4.2%)	0 (0%)	1 (8.3%)	
Complete cancer treatment received	Radiation + CCR	4 (16.0%)	1 (8.3%)	3 (23.1%)	.754
	Surgery + Radiation	11 (44.0%)	6 (50.0%)	5 (38.5%)	
	Surgery + Radiation + CCR	10 (40.0%)	5 (41.7%)	5 (38.5%)	

Abbreviation: CCR, concurrent chemoradiation.
[a]Tumor stage: AJCC eighth Edition Cancer Staging.

can inform future trials. An alpha level of .05 was used for all statistical tests for primary and secondary outcomes. All randomized participants were included in each analysis. Follow-up observations were excluded from repeated measures analysis in cases of missing data. No imputation was used to handle missing data. Participant characteristics were compared between those who did and did not complete follow-up data collection to assess possible bias introduced to study results due to participant attrition. An interim analysis was not conducted. Statistical analysis was performed using SAS version 9.4 for Windows.

Results

Baseline Characteristics

A total of 25 eligible HNC survivors completed the study. Most participants were White (n = 24, 96.0%) and male (n = 22, 88.0%) with a mean age of 61 years (S.D. = 5.7), had some college or higher education (n = 21, 84%), with a diagnosis of human papillomavirus-associated oropharyngeal cancer (n = 22, 88%), and received multi-modality cancer treatment (Table 1).

Feasibility

Recruitment. A total of 215 patients seen at Penn Medicine Head and Neck Cancer Clinics were screened for eligibility; 64 (29.8%) met the study eligibility criteria. Among eligible patients, 25 (39.1%) consented and enrolled in the study (Figure 1. CONSORT Flow Diagram). Patients (n = 39, 60.9%) declined due to: inconvenient distance to the study location (n = 18), time constraints (n = 10), lack of interest (n = 10), and COVID-19 (n = 1; Figure 1. CONSORT Flow Diagram). Recruitment was hindered by the COVID-19 epidemic. Nonetheless, we were able to enroll at the anticipated rate of 1 to 3 participants per month.

Randomization and Withdrawals. Of the 25 participants who enrolled and completed the baseline measures, 12 were

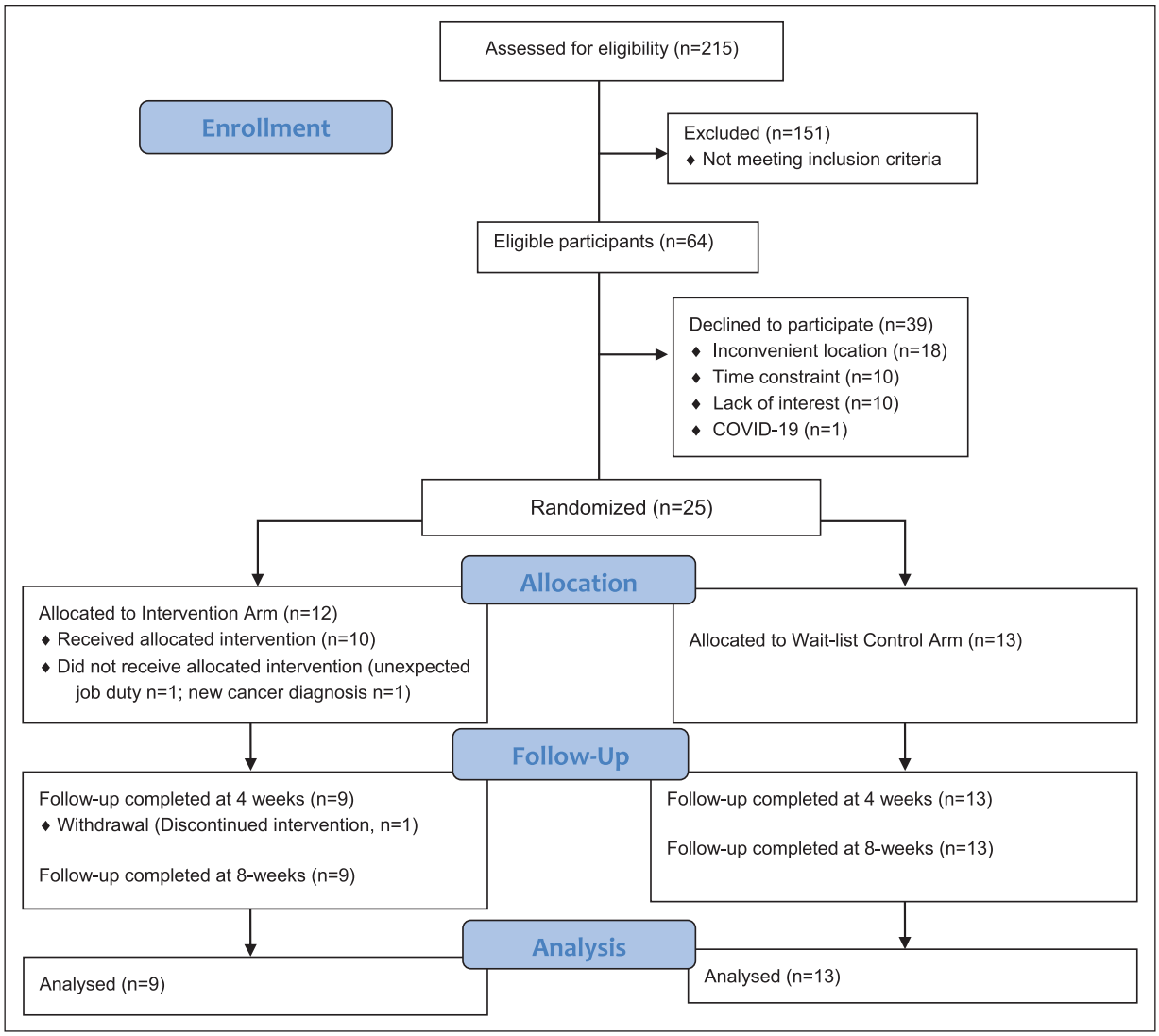


Figure 1. CONSORT flow diagram documenting the number of patients screened, consented, and withdrawn during the study assessment period.

randomly assigned to the active group, and 13 to the control group. Among the 12 participants in the active group, two participants withdrew from the study before attending the PBM therapy sessions due to unexpected job duties and a new cancer diagnosis. One participant in the active group had a history of pre-treatment skin redness from radiation dermatitis, due to impaired ability to assess for dermal toxicity, the patient was taken off the study during the intervention (see Figure 1, CONSORT Flow Diagram). No participants withdrew from the control group during the study. The attrition analysis demonstrated that no statistically significant differences were observed between participants who did and did not complete follow-up (Supplemental Table 2).

Completion Rates of PBM Therapy. Completion rate for patients participating in the study was high: 110 of 120 (91.7%) planned PBM treatment sessions (12 sessions per patient × 10 patients = 120 sessions) were completed. The reasons for missing the PBM treatment sessions included job conflict, family obligations, and COVID-19 infection.

Adverse Events. No adverse events were reported. One participant with baseline widespread pain reported temporary increased pain after the second session of PBM therapy, which resolved spontaneously and was not felt to be related to PBM treatment.

Preliminary Efficacy

External Lymphedema and Fibrosis

Individual Group Trajectory Description. All participants had at least 1 anatomical site with moderate LEF at baseline.

The active group's total severity score of external LEF at baseline was 5.4 ± 2.1 and decreased to 3.2 ± 1.9 at 8-week post-intervention. The control group's total severity score of external LEF at baseline was 4.1 ± 2.3 and increased to 4.6 ± 1.7 at 8-week post-intervention (Table 2).

The active group had 2.8 ± 0.8 anatomical sites with external LEF at baseline and 2.1 ± 0.8 sites at 8-week post-intervention. The control group had 2.5 ± 1.4 anatomical sites with external LEF at baseline and 2.8 ± 0.8 sites at 8-week post-intervention (Table 2).

Between-Group Comparison. Compared to control group participants, active group participants exhibited a significantly greater decrease in the total severity of external LEF from baseline to 8-week post-intervention. Specifically, participants in the active group were estimated to have a mean reduction in total severity of 2.9 ± 0.7 (95% CI: −4.3, −1.4) units greater than those in the control group ($P < .001$, Cohen's d [8-week post-intervention] = −1.76, Figure 2, Table 2, and Supplemental Table 3). Additionally, those in the active group exhibited a significantly greater decrease in the number of anatomical sites with LEF from baseline to 8-week post-intervention. Specifically, participants in the active group were estimated to have a mean decrease in the number of anatomical sites with LEF 1.0 ± 0.5 (95% CI: −1.9, − 0.1) units greater than those in the control group ($P = .038$, Cohen's d [8-week post-intervention] = −1.21, Figure 3, Table 2, and Supplemental Table 3).

Symptom Burden

Individual Group Trajectory Description. Symptom burden was lower at baseline in the control group (all 7 subscales' average score range between 0.6 and 1.6, Table 2). For participants in the active group, the Soft Tissue and Neurologic Toxicity subscale and the Swallowing and Taste Changes subscale reported the highest average scores at 2.5 ± 1.0 and 2.1 ± 1.5 on a 1 to 5 scale, indicating moderate levels of LEF-associated symptom burden at baseline (Table 2).

Between-Group Comparison. Participants in the active group showed improvement in symptom burden among 4 of 7 subscales of the HN-LEF Symptom Inventory compared to those in the control group from baseline to 8-week post-intervention, though not statistically significant (see Supplemental Table 3). This included the following subscales: Soft Tissue and Neurologic Toxicity, Cohen's d = −0.47 (Table 2 and Supplemental Figure 1); Body Image and Sexuality, Cohen's d = −0.34; Swallowing and Taste Changes, Cohen's d = −0.27; and Communication, Cohen's d = −0.19 (Table 2).

Neck Range of Motion

Individual Group Trajectory Description. At baseline, the active group participants' neck range of motion in degrees was 43.5 for forward flexion, 37.4 for extension, 23.3 for left lateral flexion, 27.3 for right lateral flexion, 48.8 for left lateral rotation, and 45.4 for right lateral rotation. All directions of neck range of motion in degrees were increased at 8-week post-intervention (Table 2).

At baseline, the control group participants' neck range of motion in degrees was 43.4 for forward flexion, 39.7 for extension, 25.6 for left lateral flexion, 26.6 for right lateral flexion, 46.5 for left lateral rotation, and 48.0 for right lateral rotation. Among 6 directions, 3 directions (left lateral flexion, right lateral flexion, and left lateral rotation) of neck range of motion in degrees were increased, and the rest of the directions' (forward flexion, extension, and right lateral rotation) neck range of motion in degrees were decreased at 8-week post-intervention (Table 2).

Between-Group Comparison. While not statistically significant, moderate effect sizes for an increase in the degrees of neck range of motion in the active group were observed. Specifically, extension (Cohen's d = 0.65, Supplemental Figure 2 and Table 2) and right lateral rotation (Cohen's d = 0.65, Supplemental Figure 3 and Table 2) exhibited improvement among the participants in the active group compared to those in the control group from baseline to 8-week post-intervention.

Jaw Range of Motion

Individual Group Trajectory Description. The active group's jaw range of motion in degrees was 40.2 ± 11.7 at baseline and 38.9 ± 9.5 at 8-week post-intervention. The control group's jaw range of motion in degrees was 40 ± 13.2 at baseline and 41.9 ± 10.9 at 8-week post-intervention (Table 2).

Between-Group Comparison. No significant difference was noted between the 2 groups.

Neck Disability Index (NDI)

Individual Group Trajectory Description. Active group participants' NDI scores were 21.2 ± 14.8 at baseline and 18.4 ± 14.6 at 8-week post-intervention (Table 2). Control group participants' NDI scores were 14.9 ± 11.1 at baseline and 11.9 ± 12.2 at 8-week post-intervention (Table 2).

Between-Group Comparison. Though not statistically significant, compared to participants in the control group, participants in the active group exhibited a small effect size in the decrease of the total score of NDI from baseline to 8-week post-intervention (Cohen's d = − 0.18; Table 2).

Plateau Effect. Data demonstrated improvement in outcomes plateaued after cessation of PBM therapy (Figures 2 and 3).

Discussion

Head and neck lymphedema is a common, frequently underdiagnosed and undertreated late effect of HNC therapy.

Table 2. Sample Means, Standard Deviations, and Effect Sizes for Patient Outcomes Comparing Active and Control Groups.

		Active Group			Control Group			Active vs. Control	
Measure/Tool	Variable (with units)	Baseline (n = 12; Mean ± SD)	8-wk (n = 9; Mean ± SD)	*P*-value (paired baseline vs 8-wk)	Baseline (n = 13; Mean ± SD)	8-wk (n = 13; Mean ± SD)	*P*-value (paired baseline vs 8-wk)	*P*-value – unpaired over change (Baseline/8-wk)	Cohen's *d*
Head and Neck – External LEF Assessment Criteria	Total Severity of External LEF (score)	5.4 ± 2.1	3.2 ± 1.9	.001	4.1 ± 2.3	4.6 ± 1.7	.327	<.001	−1.76
	Number of Sites with External LEF (score)	2.8 ± 0.8	2.1 ± 0.8	.003	2.5 ± 1.4	2.8 ± 0.8	.535	.038	−1.21
Head and Neck – LEF Symptom Inventory	Soft Tissues and Neurologic Toxicity (score)	2.5 ± 1.0	1.7 ± 1.3	.018	1.6 ± 1.0	1.2 ± 0.9	.056	.289	−0.47
	Body Image and Sexuality (score)	1.1 ± 1.4	0.8 ± 1.3	.046	0.6 ± 0.9	0.4 ± 0.5	.370	.544	−0.34
	Swallowing and Taste Changes (score)	2.1 ± 1.5	1.5 ± 1.2	.056	1.6 ± 0.8	1.3 ± 1.0	.278	.560	−0.27
	Communication (score)	1.4 ± 1.0	0.8 ± 0.9	.148	1.4 ± 1.3	1.0 ± 1.1	.091	.656	−0.19
	Mucosal Irritation (score)	0.7 ± 0.8	0.7 ± 0.9	.594	1.1 ± 0.8	0.8 ± 0.8	.128	.432	0.28
	Jaw and Oral Dysfunction (score)	1.9 ± 1.5	1.7 ± 1.2	.108	1.5 ± 1.1	1.0 ± 0.9	.032	.495	0.20
	Systemic Symptoms and Social Functioning (score)	1.9 ± 1.5	1.4 ± 1.3	.160	1.1 ± 1.0	0.7 ± 0.5	.056	.954	0.05
Cervical Range of Motion Device	Forward flexion (degree)	43.5 ± 8.3	46.6 ± 15.3	.430	43.4 ± 5.6	42.7 ± 7.1	.707	.319	0.44
	Extension (degree)	37.4 ± 13.0	44.1 ± 11.3	.138	39.7 ± 13.8	39.6 ± 10.4	.957	.112	0.65
	Left lateral flexion (degree)	23.3 ± 9.3	29.2 ± 10.4	.014	25.6 ± 10.9	28.6 ± 7.7	.150	.451	0.25
	Right lateral flexion (degree)	27.3 ± 8.9	30.0 ± 8.5	.280	26.6 ± 6.5	29.3 ± 10.1	.385	.985	0.01
	Left lateral rotation (degree)	48.8 ± 15.4	52.1 ± 11.7	.517	46.5 ± 9.6	54.1 ± 9.5	.001	.164	−0.67
	Right lateral rotation (degree)	45.4 ± 13.5	52.8 ± 11.2	.044	48.0 ± 7.0	47.9 ± 10.9	.947	.122	0.65
Jaw Range of Motion Scale	Jaw range of motion (degree)	40.2 ± 11.7	38.9 ± 9.5	.791	40.0 ± 13.2	41.9 ± 10.9	.251	.398	−0.32
Neck Disability Index	Neck Disability Index (Score)	21.2 ± 14.8	18.4 ± 14.6	.096	14.9 ± 11.1	11.9 ± 12.2	.181	.741	−0.18

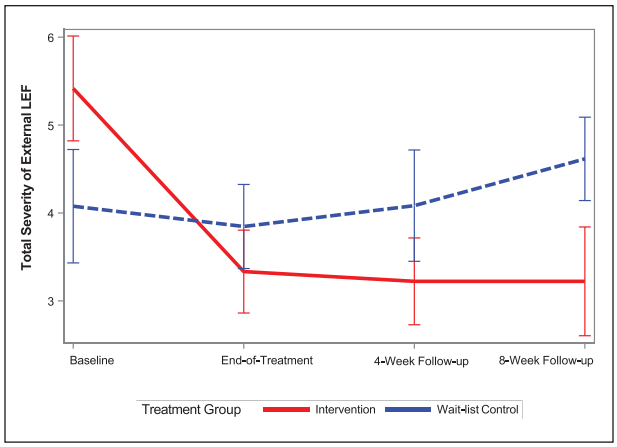


Figure 2. Total Severity of External LEF (mean ± standard error). This shows the mean of the total severity score of external LEF from baseline to 8 weeks for patients in the intervention group (red line) and the control group (blue line) with standard errors.

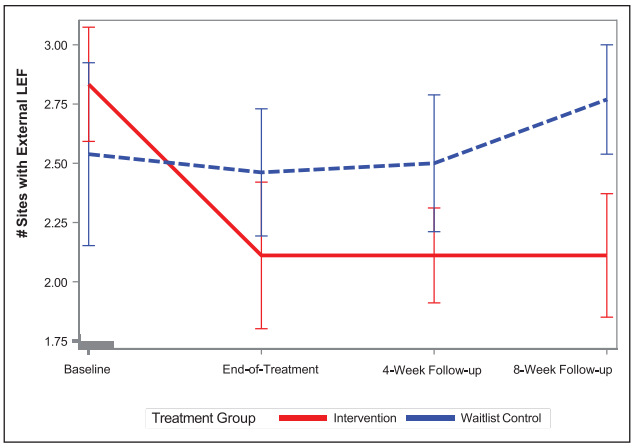


Figure 3. # Sites with External LEF (mean ± standard error). This shows the mean number of anatomical sites with external LEF from baseline to 8-week for patients in the intervention group (red line) and the control group (blue line) with standard errors.

Timely referral and early initiation of complete decongestive therapy, which is standard care for lymphedema, is critical to decrease swelling, address related symptom burden and functional impairment, and reduce the healthcare cost associated with its sequelae. Without timely and adequate treatment, head and neck lymphedema can progress over time and become a chronic condition that is challenging to treat and refractory to standard therapy. In recent years, several alternative and complementary strategies [eg, surgical approaches (liposuction),[43] Kinesio taping,[44] and pneumatic compression device[45]] have been reported to manage head and neck lymphedema. Most of them focused on early-stage head and neck lymphedema. A study targeted at addressing chronic soft tissue damage from HNC treatment showed that a tailored yoga program may improve symptom burden and shoulder function in HNC survivors.[46] Nevertheless, there are no proven effective therapies for treating chronic or refractory head and neck lymphedema. Thus, novel treatment modalities are needed in this area.

Head and neck cancer survivors with chronic or refractory lymphedema are often overwhelmed by the complexity of their daily self-care regimen. Furthermore, they are at risk for worsening neuromusculoskeletal toxicity if therapeutic interventions are not carefully administered.[14] Thus, the safety and feasibility of novel therapeutic modalities are of primary concern. Our prior single-arm clinical trial indicated that PBM therapy was feasible and potentially effective for treating head and neck chronic lymphedema.[31] The results from this pilot RCT further support that PBM is a safe and feasible intervention for HNC survivors with chronic or refractory lymphedema. No adverse events were reported by participants during the study, and preliminary data indicates potential efficacy.

Compliance with self-care and therapy is highly variable in the HNC survivor population. Barriers to care have been previously described.[47] Of note, data indicates that many patients discontinue lymphedema therapy before completing the planned intervention.[47] Patients who have benefited from therapy may be more willing to continue treatment or try alternative therapeutic modalities for recurrent or residual disease. Conversely, patients who experienced barriers to lymphedema care or those who have had a poor response to therapy may be less willing to comply fully with planned therapy or to try alternative therapeutic regimens. Therapeutic regimens that require frequent visits may be particularly challenging for this group of patients.[48] Patients in this study had previously received or been offered participation in lymphedema therapy (ie, complete decongestive therapy) with mixed therapeutic results. Nonetheless, the completion rate of the planned PBM therapy for patients participating in this trial was high at 92%. The high completion rate for both groups in this study may speak to the high symptom burden of chronic lymphedema and the strong motivation of this patient population to find effective therapeutic solutions and perceived benefits.[48]

Our data demonstrated improvement in outcomes for patients randomized to the intervention group compared to the wait-list control group at 8-week post-intervention. This was noted across the clinician-reported outcome measure (CRO), objective measures, and subjective patient report. Using the HN-LEF Assessment Criteria,[34,35] a clinical measure of soft tissue swelling and firmness, we demonstrated that participants randomized to the intervention group showed a statistically significant decrease in the total severity of external LEF and the number of anatomical sites with LEF. The Cohen's d value of 1.76 represents the effect of the intervention group decreasing in total severity by

2.98 units more than the control group from baseline to the 8-week timepoint. The changes observed in the total severity of external LEF and the number of anatomical sites with LEF indicate a clinically significant improvement.

We also noted moderate effect sizes for a trend toward improvement in the degrees of neck range of motion, including extension and right lateral rotation. These were surprising findings given the small number of participants and the fact that these patients had chronic and/or refractory lymphedema.

Our data also suggests that PBM may improve symptom burden. Patients treated with PBM showed improvement on 4 of 7 subscales of the HN-LEF Symptom Inventory, including the following: Soft Tissue and Neurologic Toxicity, Body Image and Sexuality, Swallowing and Taste Changes, and Communication. The Soft Tissue and Neurologic Toxicity Subscale, which exhibited the largest effect size of all the HN-LEF Symptom Inventory subscales, consists of items such as "feeling uncomfortable in one's head or neck", "tightness," "stiffness," "numbness," "tenderness," and "limited head and neck movement," which drives the positive findings. Although these findings were not statistically significant, we believe they are suggestive of benefit and worthy of further investigation in a larger trial.

Of note, our data demonstrated improvement in outcomes, which plateaued after cessation of PBM therapy. This leads to several important questions. Would a more prolonged course of treatment maintain the observed benefit? Alternatively, would an intermittent maintenance approach be more effective and feasible? Future studies with larger sample sizes will be needed to confirm the results of our pilot trial and address the questions of schedule and duration of PBM therapy.

The participants in this trial had an average of 2 to 3 anatomical sites involved with LEF and at least 1 site with moderate LEF at baseline. In addition, the participants had moderate levels of symptom burden with 2 subscale scores of 2 to 2.5 on a scale of 1 to 5 at baseline. In the survivor population, this level of chronic LEF and symptom burden is high and clinically significant. Patients with this degree of symptom burden require intervention. Unfortunately, patients in our study had previously failed standard therapeutic interventions. Thus, the positive initial findings noted in this pilot study are encouraging.

Limitations and Strengths

Limitations of this study include the small sample size, potential selection bias, potential baseline imbalance, lack of blinding, and lack of long-term follow-up. Lack of blinding may bias patient-reported outcomes and clinician-reported outcome measures. Our ongoing triple-blind randomized trial addresses this limitation.[49] Despite randomization, baseline imbalances between the groups were noted. Although the 2 groups were comparable in their demographic variables, participants randomized to the intervention group had higher LEF-related burden. The issue of LEF imbalance may be addressed by definitive randomized phase III trials that are adequately powered. Also, the COVID-19 restrictions impacted the implementation of the study protocol in multiple ways, including recruitment, intervention delivery, and data collection, particularly internal lymphedema data. Some eligible patients declined participation due to COVID-19-related travel restrictions. In addition, COVID-19 transmission occurs through both droplets and aerosols.[50] Endoscopic procedures carry a potentially high risk of transmission due to close patient contact, aerosol generation, and surface contamination. Thus, it was impossible to collect data on internal lymphedema, and we did not know the effect of PBM therapy on internal lymphedema in this study. This impacts the generalizability of the study findings to HNC survivors with internal lymphedema. Due to the funding limits and the study's duration, we could not collect data beyond 8 weeks. This limited our ability to evaluate the durability of treatment effects for chronic lymphedema in the study population. This limitation is being addressed in our ongoing randomized trial, which will include a 12-month follow-up.[49] Finally, we did not specifically assess the impact of PBM on fibrosis. The strengths of this pilot study include the randomized study design and the use of a comprehensive set of assessment tools, including clinician-assessed, objective, and subjective measures.

Conclusions

PBM is both safe and feasible and may improve chronic lymphedema in HNC survivors. Large randomized controlled trials are needed to confirm these clinical findings and establish the optimal therapy schedule.

Author Note

Lucy P. Andersen is currently employed by Johnson & Johnson Innovative Medicine, and her contribution to this research occurred during her time as a PhD student at the University of Pennsylvania.

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Ethical Considerations

The study was approved by the Institutional Review Board at the study site.

Consent to Participate

All participants signed informed consent forms.

Author Contributions

Jie Deng contributed to the conception and design of the study. Material preparation and data collection were performed by Alexander Lin, John N. Lukens, Erin McMenamin, Karthik Rajasekaran, Joy C. Cohn, Wendy Schlessinger, and Lucy P. Andersen. Barbara Murphy provided critical feedback on the analysis and interpretation of results. Bryan A. Spinelli provided critical input on the interpretation of results. Ryan J. Quinn performed data analysis. Jie Deng wrote the first draft of the manuscript, and all authors commented on previous versions. All authors reviewed and approved the final manuscript.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supplemental Material

Supplemental material for this article is available online.

Trial Registration Number and Date of Registration

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